

Clinical Characterization of an Inherited Disorder Associated with Hyperactivity and Gait Impairment

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The parents are first cousins (paternal uncle's son and paternal uncle's daughter), and have two affected daughters with intellectual disability and five unaffected daughters. A positive family history is evident in the pedigree. The affected individuals' paternal uncle reportedly has similar clinical features and has one affected son, as well as one unaffected daughter and one unaffected son. The affected individuals exhibit severe intellectual disability with markedly limited ability to acquire academic skills, along with aggressive behavior, hyperactivity, restlessness, delayed gait acquisition, and impaired balance. At the time of clinical evaluation, the probands were 37 and 28 years old, respectively. Their head circumferences were 57 cm (+2.42 SD) and 56 cm (+1.52 SD), respectively.

The patient's NGS data were initially filtered based on population frequency databases, including ESP6500¹, ExAC², and the 1000 Genomes³ Project. Synonymous variants were subsequently excluded, and exonic, splicing, and exonic;splicing variants were retained. Based on the pedigree analysis, which indicated an autosomal recessive (AR) inheritance pattern, homozygous variants were retained for further analysis. *LRRC40*⁴ and *FRS3* were ultimately identified as candidate genes based on evaluation of the patient's phenotype and comparison with previously published reports.

Genetic analysis identified two homozygous nonsynonymous variants in the *LRRC40* and *FRS3* genes. A variant in *LRRC40*, located on chromosome 1 at position 70,611,496, was identified as a C>A substitution. The variant is located in an exonic region and results in a nonsynonymous amino acid change. In silico prediction analyses classified this variant as deleterious by both SIFT and PolyPhen with a CADD score of 35. According to the Iranome database, the variant has a reported allele frequency of 0.000416 and has been observed in a single heterozygous individual of Baluchi ancestry.

A second nonsynonymous variant was identified in *FRS3* on chromosome 6 at position 41,739,633, corresponding to a C>A substitution within an exonic region. SIFT predicted this variant to be deleterious with a CADD score of 23. Notably, this variant has not been reported in the Iranome database to date, suggesting that it may be a rare variant in the Iranian population.

LRRC40 encodes leucine-rich repeat-containing protein 40, whose biological function has not yet been fully elucidated. Given the structural properties of leucine-rich repeat proteins, *LRRC40* may facilitate protein–protein interactions and participate in diverse cellular signaling processes⁴. *LRRC40* has been identified as a component of the LRIG1-associated protein interaction network, where it may influence receptor tyrosine kinase signaling and potentially contribute to tumor suppressive mechanisms⁵. Of particular relevance to neurodevelopment, a homozygous missense variant in *LRRC40* was reported in an

individual with autism spectrum disorder. Functional analyses demonstrated substantially reduced LRRC40 protein expression and identified an interaction between LRRC40 and Plexin A2, supporting a possible role for LRRC40 in neural signaling and developmental processes⁴. In addition, LRRC40 has been implicated in cancer-related prognostic signatures, including those associated with Wilms tumor⁶. Transcriptomic studies of tissue regeneration have also identified LRRC40 among genes responsive to injury and immune-related signaling, suggesting that its biological functions may involve multiple cellular pathways⁷.

FRS3 encodes fibroblast growth factor receptor substrate 3 (FRS3), an intracellular adaptor protein also referred to as FRS2 β and SUC1-associated neurotrophic factor target 2 (SNT-2). This protein functions in both fibroblast growth factor (FGF) and neurotrophic signaling pathways, which are essential for the development and maintenance of the nervous system and contribute to processes such as neural cell proliferation, differentiation, and synaptic plasticity. Beyond its adaptor function, FRS3 has been reported to interact with microtubules and lysosomes in neuronal cells, suggesting that it may also contribute to microtubule organization and stability, axonal transport, and intracellular lysosomal degradation⁸. FRS3 has a potential role in neurodevelopment based on its expression in developing neurons and functional studies demonstrating involvement in neuronal maturation⁹⁻¹¹.

The identified variants in LRRC40 and FRS3 may be relevant to the patient's neurological phenotype, particularly Hyperactivity and Gait Impairment. LRRC40 has been previously implicated in neurodevelopmental disorders and reported in association with ADHD and hyperactivity⁴.

FRS3 has functional evidence supporting a role in neuronal development and neural circuitry, processes that are essential for the proper development and coordination of motor pathways^{9,12}. Therefore, disruption of FRS3-related neuronal development¹² may provide a plausible biological basis for Gait Impairment, although direct clinical evidence remains limited.

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